MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8169

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Tetrode

See figure 1

Mounting position: Vertical, base down or up

Weight: 5.5 pounds nominal

ABSOLUTE RATINGS: F1 = 150 MHz

Parameter: Unit:	Ef Vac	Eb kVdc	Ec2 Vdc	Ec1 Vdc	Ib Adc	Pg1 W	Pg2 W	Pp kW	T(anode core & seal) °C	Cooling --- Note 2
Maximum:										
Class C Teleg:	9.0±5%	7.0	1,000	-500	2.0	50	175	3.0	250	---
Class C Telep: (anode mod)	9.0±5%	5.0	600	-500	1.4	50	175	2.0	250	---
Class AB1:	9.0±5%	7.0	1,000	---	2.0	50	175	3.5 Note 3	250	---
TEST CONDITIONS:	9.0	2.0	750	Adj	1.0	---	---	---	---	Note 4

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1231	Peak emission	eb = ec2 = ec1 = 2.5 kv	0.65	II	is	40	---	a
1301	Filament current		0.65	II	If	39.5	43.5	Aac
1261	Electrode voltage (grid)		0.65	II	Ec1	-95	-127	Vdc
1266	Total grid current		0.65	II	Ic1	---	-20	μAdc
1256	Electrode current (screen)		0.65	II	Ic2	0	+25	mAdc
1266	Primary-grid emission (control)	Ef = 10 V; Ic1 = 400 mAdc; t = 15; anode and g2 grounded	0.65	II	Isg1	---	-50	μAdc
1266	Primary-grid emission (screen)	Ef = 10 V; Ic2 = 400 mAdc; t = 15; Ec1 = 0; anode grounded	0.65	II	Isg2	---	-100	μAdc
	Quality conformance inspection, part 2							
1372	Current division (method B, short pulse)	Eb = Ec2 = 850 Vdc Ec1 = -300 Vdc; egk/ib = 5.5 a	---	---	egk ic2	--- ---	0 0.95	v a
1261	Electrode voltage (grid)	Eb = 4.0 kVdc; Ec2 = 850 Vdc; Ec1/Ib = 1.0 mAdc	---	---	Eco	---	-310	Vdc
1331	Direct-interelectrode capacitance	Grounded cathode	---	---	Cin Cout Cgp	120 10.5 ---	140 14.5 1.40	pF pF pF
1331	Direct-interelectrode capacitance	Grounded grid	---	---	Cin Cout Cpk	55.0 10.5 ---	67.0 14.5 0.20	pF pF pF
2204	Linear amplifier power output and distortion, two-tone method	Class AB1 amplifier; F = 2 to 30 MHz; Eb = 5.0 kVdc; Ec2 = 850 Vdc; Ec1/Ibo = 500 mAdc; Eg1/Po = 5.0 kW (useful); Rf = 1,700 ±100 ohms; anode loaded Q = 15 ±2; Rg = 1,000 ohms (max)	---	---	3rdIM 5thIM	-32 -37	--- ---	dB dB
	Quality conformance inspection, part 3							
---	Service-life guarantee	See note 1	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Periodic-check tests							
1042	Shock, specified pulse	No voltages applied; accel = 15 G peak (min); shock = 11 ms half-sine; total impact = 6 (3 each X and Z axes)	---	---	---	---	---	---
---	Shock, specified pulse end points:							
1261	Electrode voltage (grid)		---	---	Ec1	-95	-127	Vdc
1266	Total grid current		---	---	Ic1	---	-25	μAdc
1032	Vibration, mechanical	No voltages applied; accel = 2 G peak (min); F = 10 to 50 Hz, ascending only; sweep t = 3 to 8 minutes; 1 sweep each X and Z axes	---	---	---	---	---	---
---	Vibration, mechanical end points:							
1261	Electrode voltage (grid)		---	---	Ec1	-95	-127	Vdc
1266	Total grid current		---	---	Ic1	---	-25	μAdc

NOTES:

1. The tube manufacturer warrants the tube for one year from the date of shipment, or 1,000 hours of filament life, whichever first elapses. This warranty applies only when the tube is operated within the maximum ratings (see "Absolute ratings" of MIL-E-1). A defective tube shall either be replaced, or at the option of the manufacturer a credit shall be made in the amount of the original purchase price prorated on the basis of 1,000 hours of "filament-on" time.

2. Minimum airflow requirements for incoming air at 40° C maximum at sea level, for operation below 30 MHz, are shown in table I. Additional cooling may be required for operation above 30 MHz. In all applications, an air-system socket, such as the EIMAC SK-1400 with air chimney such as the EIMAC SK-1406, or equivalents, should be used. Air should be directed in a base-to-anode direction, and the values shown are for the tube and the SK-1400/SK-1406 combination with the air so directed. In all cases, enough air should be used to prevent operation with the anode core and/or seal temperatures above the maximum rating, and where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial. Air should be applied before or simultaneously with the application of filament voltage, and may be removed simultaneously with the electrode voltages.

TABLE I. Minimum airflow requirements.

Anode dissipation (watts)	Airflow (cfm)	Approximate pressure drop (in. H_2O)
1,500	36.5	0.3
2,500	60	0.8
3,500	86	1.6

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NOTES: -Continued

3. During continuous Class AB1 operation, it is not likely that the absolute maximum ratings for the anode voltage (Eb) and anode current (Ib) can be used simultaneously, as efficiency would have to be very high (75 percent) in order to avoid exceeding the anode dissipation (Pp) rating of 3.5 kW.

4. In all electrical tests involving application of filament voltage, an air-system socket and chimney may be used and forced-air cooling is permissible.

Custodians:
Army - EL
Air Force - 80

Review activities:
Army - MU
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2578)

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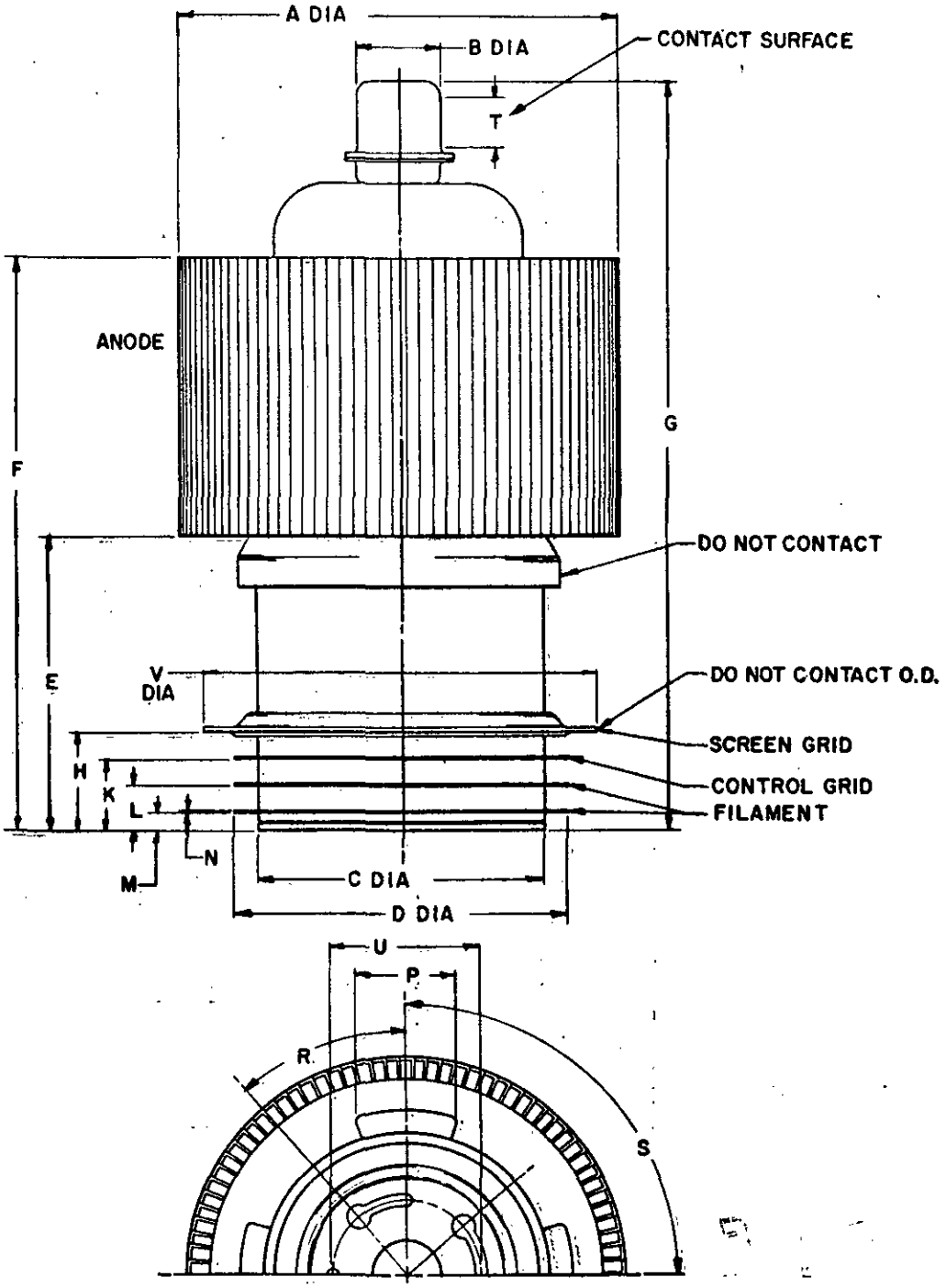


FIGURE 1. Outline drawing of electron tube type 8169.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	4.560 (115.82)	4.625 (117.48)
D	3.490 (88.65)	3.550 (90.17)
E	2.875 (73.03)	3.125 (79.38)
H	.965 (24.51)	1.005 (25.53)
K	.700 (17.78)	.730 (18.54)
L	.430 (10.92)	.460 (11.68)
M	.160 (4.06)	.180 (4.57)
N	.018 (.46)	.025 (.64)
P	1.050 (26.67)	1.150 (29.21)
R	39°	41°
S	89°	91°
U	1.557 (39.55)	1.567 (39.80)
Quality conformance inspection, part 3 (periodic check)		
B	.860 (21.84)	.890 (22.61)
C		3.050 (77.47)
F	5.875 (149.23)	6.125 (155.58)
G	7.600 (193.04)	7.900 (200.66)
T	.485 (12.32)	
V	4.000 (101.60)	4.175 (106.05)

FIGURE 1. Outline drawing of electron tube type 8169 - Continued.

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